PHASE 8 · ADVANCED PATTERNS & CROSS-CLUSTER OPERATIONS

Chapter 28

Advanced Patterns: Multi-Region, Cross-Cluster & Cost Optimization



Global scale · disaster recovery · zero-trust federation

6–7 hrs

CHAPTER GOAL

Deploy for Global Scale, DR & Cost Efficiency

Scale the DevSecOps platform to multiple data centers and cloud regions — with automatic failover, geo-aware traffic routing, and right-sized spend.

Multi-region cluster architecture

Active-active deployments across geographies

Cross-region disaster recovery

Automated failover with sub-hour RPO

Global load balancing

Geo-DNS and service-mesh traffic routing

Cost optimization strategies

Right-sizing, spot instances, tiered infra

Zero-trust federation

Authenticate & authorize every connection

Multi-cloud deployments

KubeFed across AWS, GCP, and Azure

From Single Point of Failure to Active-Active

SINGLE REGION (CURRENT)

3 Control Planes (HA etcd)

2+ Worker Nodes

Applications & Services

Single point of failure — region down = complete outage

MULTI-REGION ACTIVE-ACTIVE

REGION 1
US East

REGION 2
EU Central

Global Load Balancer — geo / latency split

- No single-region SPOF
- Low latency (geo-distributed)
- Automatic failover in seconds
- Multi-region high availability

KubeFed: One Deployment, Many Clusters

1

Install KubeFed

Helm-install on the host cluster (Region 1), then repeat on every member cluster.

2

Define FederatedDeployment

Set replica placement per cluster — e.g. 3 replicas in US-East, 2 in EU-Central.

3

Federated DNS

CoreDNS routes myapp.svc.cluster.local automatically to the nearest cluster.

FederatedDeployment: myapp-global

cluster-us-east

replicas: 3

cluster-eu-central

replicas: 2

```
image: harbor.devsecops.com/myapp:latest
```

Route Users to the Best Region, Automatically

Geo-DNS Routing

Users in US	→ **Region 1 (US)**
Users in EU	→ **Region 2 (EU)**
Everyone else	→ **Primary region**

Policy: latency-based routing resolves to the lowest-latency region, with automatic fallback on failure.

Service Mesh Global Routing (Istio)

`myapp.global.svc.cluster.local`

Region 1 endpoint	Region 2 endpoint
50%	**50%**

Automatic failover if Region 1 goes down — VirtualService detects & reroutes traffic.

Active-Passive Failover

REGION 1 — PRIMARY

- All traffic
- Active apps
- Live databases

RTO: 30–60 min

REGION 2 — STANDBY

- Minimal traffic
- Standby apps
- Replicated data

RPO: < 1 hour

⇄ *continuous async replication*

Monitor health → **Region 1 fails** → **Switch DNS → Region 2** → **Promote to primary**

Keep Data in Sync — and Recoverable

PostgreSQL Streaming Replication

Logical replication — crosses versions and platforms

```
Region 1 publishes: CREATE PUBLICATION myapp_pub
```

```
Region 2 subscribes: CREATE SUBSCRIPTION myapp_sub
```

- All changes replicated in real time
- Read replicas from Region 2
- Zero-downtime promotion if needed

Backup Schedule

Every 6 hrs	Full etcd backup → S3 (different region)
Every 24 hrs	PostgreSQL + app state → cold storage (Glacier)
Weekly	Test recovery procedure & verify data integrity
Monthly	Retention, encryption & compliance audit

Right-Size Resources, Cut Waste

BEFORE — Overprovisioned

Pod requests: 500m CPU · 512Mi RAM

Actual usage: 100m CPU · 128Mi RAM

80% over-provisioned

AFTER — Right-Sized

Pod requests: 150m CPU · 200Mi RAM

Actual usage: 100m CPU · 128Mi RAM

60% less cloud spend

Spot Instances & Preemptible VMs

Use cheap, interruptible capacity for non-critical batch & CI/CD workloads.

```
tolerationSeconds: 30
```

graceful shutdown before preemption

Match Workload Criticality to Instance Tier

Tier 1 · Premium

On-Demand VMs

Control plane · stateful databases

~$0.50/hr

99.99% availability

Tier 2 · Standard

Reserved Instances

Production workloads, 1–3 yr commitment

~$0.25/hr (–50%)

99.9% SLA

Tier 3 · Spot

Preemptible VMs

Batch jobs, CI/CD, dev/test

~$0.05/hr (–80%)

95% best-effort

~~$1,000~~
BEFORE / month

→

$300
AFTER / month

70%

Overall cost reduction achieved by matching workload criticality to the right pricing tier.

Zero-Trust Federation

Policy: every access must be authenticated & authorized — even within the same cluster or region.

BEFORE — Trust at Boundary

- Pod-to-pod: TLS
- Service-to-DB: no auth (implicit trust)

Result: a compromised pod can freely reach the database.

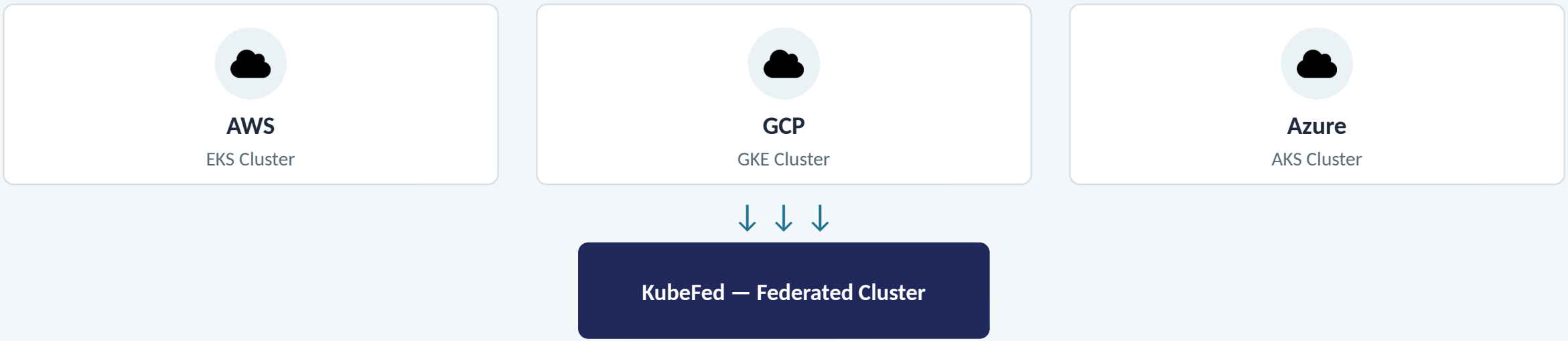
AFTER — Zero-Trust

- Authenticated (certificate)
- Authorized (policy)
- Encrypted (mTLS)
- Logged (audit)

Every pod-to-pod, pod-to-DB, and pod-to-API call is verified and logged.

Microsegmentation: default-deny NetworkPolicy blocks all traffic; explicit policies allow only app → database on port 5432.

One Chart, Three Clouds — No Vendor Lock-In



- Same apps across all clouds
- Kubernetes-native — cloud-agnostic
- Avoid vendor lock-in

```
helm install myapp ./myapp-chart -f values-aws.yaml
helm install myapp ./myapp-chart -f values-gcp.yaml
helm install myapp ./myapp-chart -f values-azure.yaml
```

Keep Data — and Audit Trails — Where They Belong

Data Residency (GDPR example)

Primary	EU — Germany
Secondary	EU — Ireland (backup only)
Tertiary	EU — Amsterdam (cold storage)

NO data leaves the EU to US or non-EU regions

Enforced via data classification in Vault, storage-class selection by region, and network policy.

Centralized Audit Logging

All regions

Elasticsearch (EU-only)

Kibana (searchable, auditable)

Monthly compliance report

Certified: GDPR · DPA · HIPAA

Failover Procedure — Region 1 Unavailable

1 — Detect

1 min

Alerting flags all nodes down; health checks fail across the board.

→

2 — Assess

2 min

Check connectivity & cloud provider status; confirm Region 1 is down.

→

3 — Activate

3 min

Switch DNS to Region 2; promote it from standby to primary.

→

4 — Restore

5–30 min

Verify services operational; restore from backup if needed.

→

5 — Notify

30 min

Notify stakeholders, update status page, begin post-incident review.

TOTAL RTO: < 5 minutes

TOTAL RPO: < 1 hour

Think Global, Act Local

Pattern	Purpose	Use Case
Federation	Unified management	Multi-cluster
Geo-DNS	Geographic routing	Global users
Service Mesh	Global traffic	Microservices
Replication	Data sync	Disaster recovery
Spot Instances	Cost saving	Non-critical work
Network Policy	Zero-trust	Security
Multi-Cloud	Avoid lock-in	Vendor flexibility

BEFORE Multi-Region

- ✕ One region fails = complete outage
- ✕ High latency for distant users
- ✕ Over-provisioned everywhere
- ✕ Hard to meet data residency

AFTER Multi-Region

- ✓ Failover in seconds
- ✓ Low latency (geo-distributed)
- ✓ Cost-optimized, right-sized
- ✓ Compliant — data stays put
- ✓ Scalable — add regions as needed